

Project 6- Student Performance Dashboard

Goal:

Track students' scores, compare subject-wise performance, calculate average marks, and highlight top/bottom scorers.

STEP 1: Open Excel

- Launch **Microsoft Excel**
- Create a **new blank workbook**

STEP 2: Enter Student Data in Excel

Type the following into your Excel sheet starting from **cell A1**:

Name	Subject	Marks
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Alice	Math	85
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Alice	Science	78
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Bob	Math	92
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Bob	Science	81
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Charlie	Math	68
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Charlie	Science	74
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David	Math	88
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David	Science	90
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STEP 3: Save the Excel File

Follow these steps:

1. Click on **File** (top-left in Excel)
2. Select **Save As** (or **Browse**)
3. Choose a folder (e.g., **Desktop** or **Documents**)

4. In the **File name** box, type:
StudentPerformance.xlsx
5. In **Save as type**, make sure it's set to:
Excel Workbook (*.xlsx)
6. Click **Save**

Your Excel file is now ready for import into Power BI.

STEP 4: Load the Data into Power BI

Follow these steps:

1. Open **Power BI Desktop**
2. In the top menu, click **Home > Get Data > Excel**
3. In the file browser window:
 - ❖ Navigate to where you saved StudentPerformance.xlsx
 - ❖ Select the file and click **Open**
4. In the **Navigator**:
 - ❖ Tick the checkbox next to the sheet/table (you'll see a preview)
 - ❖ Click the **Load** button

Your student data is now loaded into Power BI!

STEP 5: Create a Bar Chart – Average Marks per Student

Follow these steps:

1. In Power BI, go to the **Report View** (white canvas)
2. In the **Visualizations** panel, click on the **Clustered Column Chart** icon (vertical bars)
3. A blank chart appears on the canvas

Add fields to the chart:

- ❖ Drag **Name** into the **X-Axis (Axis)**
 - ❖ Drag **Marks** into the **Y-Axis (Values)**
 - ❖ By default, it shows **Sum** of Marks
 - ❖ Click the dropdown next to **Marks** in the **Values** area
 - ❖ Select **Average**
4. Click the chart title and change it to:
"Average Marks per Student"

You now have a chart showing how well each student is performing on average.

STEP 6: Create a Bar Chart – Subject-wise Marks Comparison

Follow these steps:

1. Click on a **blank area** of the canvas (so you don't edit the first chart).
2. In the **Visualizations** panel, click on the **Stacked Bar Chart** (horizontal bars icon)
3. A new blank chart appears.

Add these fields:

- ❖ Drag **Subject** into the **Y-Axis**
 - ❖ Drag **Marks** into the **X-Axis (Values)**
 - ❖ Drag **Name** into the **Legend**
4. Click the chart title and rename it to:
"Subject-wise Marks by Student"

Now you'll see a colored bar for each student under each subject — great for comparing performance across subjects!

STEP 7: Create KPI Cards – Highest and Lowest Marks

A. Create the Highest Mark Card

1. Click on a **blank area** of the canvas.
2. In the **Visualizations** panel, click the **Card** icon ( single number display).
3. A blank card will appear.
4. In the **Fields** panel:
 - ❖ Drag **Marks** into the **Values** section.
 - ❖ By default, it shows **Sum**.
→ Click the dropdown next to **Marks** and select **Maximum**
5. Click the card title, go to the **Format (paint roller)** panel, expand **Title**, and rename it to:
"Highest Mark"

B. Create the Lowest Mark Card

1. Click on another blank area of the canvas.
2. Insert another **Card**.
3. Drag **Marks** into **Values** again.
4. Click the dropdown and select **Minimum**
5. Rename the title to:
"Lowest Mark"

You now have summary KPI cards showing top and bottom scores.

STEP 8: Add Table View and Slicer

A. Create a Table to View All Data

1. Click on a **blank area** of the canvas.
2. In the **Visualizations** panel, click the **Table** icon (grid-style).
3. In the **Fields** panel:

Drag **Name**, **Subject**, and **Marks** into the table.

This gives you a clear view of all raw data — useful for checking individual subject scores.

B. Add a Slicer to Filter by Name

1. Click on a blank space again.
2. In the **Visualizations** panel, click the **Slicer** icon (filter icon).
3. Drag the **Name** field into the slicer.

Now, when you click a name in the slicer, all visuals will automatically update to show that student's performance only.

STEP 9: Format and Style the Dashboard

A. Rename All Visual Titles

1. Click each visual (chart, card, table).
2. Go to the **Format (paint roller)** panel.
3. Expand **Title** → Make sure it's **ON**.
4. Change titles to clear, readable names like:
 - ❖ "Average Marks per Student"
 - ❖ "Subject-wise Marks by Student"
 - ❖ "Highest Mark"
 - ❖ "Lowest Mark"
 - ❖ "Student Records Table"
 - ❖ "Filter by Name"

B. Turn on Data Labels (for bar charts)

1. Click on each **bar chart**.
2. Go to **Format > Data Labels**
3. Toggle **ON** — this shows actual mark values on each bar.

C. Apply a Dashboard Theme

1. Click the **View** tab in the top menu.
2. Choose a theme like:

- ❖ "Colorblind Safe"
- ❖ "Minimal"
- ❖ "Classic"

Themes apply consistent colors and fonts across your visuals.

D. Align & Organize Visuals

1. Arrange visuals in a neat grid.
2. Place **cards** at the top.
3. Keep **charts** in the middle.
4. Place the **table and slicer** on the side or bottom.

Use:

Snap to grid and **Gridlines** from **View** menu for clean alignment.

STEP 10: Save and Export the Dashboard

A. Save Your Power BI Project File

1. Click on **File** (top-left corner)
2. Choose **Save As**
3. Choose a folder (e.g., **Desktop** or **Documents**)
4. In the **File name** box, type:
`StudentPerformanceDashboard.pbix`
5. Click **Save**

This .pbix file saves all your work and can be reopened anytime in Power BI Desktop.

B. Export as PDF (Optional)

1. Click **File > Export > Export to PDF**
2. Power BI will generate a PDF version of your dashboard
3. Save the PDF to share with others offline

C. Publish to Power BI Service (Optional)

1. Click **Home > Publish**
2. Sign in with your Microsoft account
3. Select **My Workspace**
4. Your dashboard is now hosted online — share it with others via a link

